

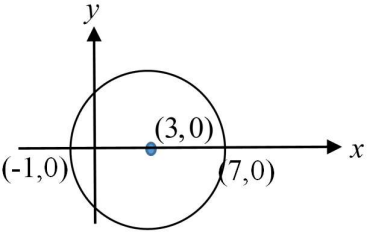
2020 RI H2 Mathematics Prelim Paper 1 Solutions

1	A particle is moving along the curve with equation $16x^2 + 9y^2 = 144$. $\frac{dy}{dt} = 2 \text{ cm s}^{-1}$ and $\frac{dx}{dt}$ is positive when $x = \sqrt{5}$. Find the rate of increase of x at this instant. [4]
[4]	$16x^2 + 9y^2 = 144$ When $x = \sqrt{5}$: $y = \pm \sqrt{\frac{144 - 16(5)}{9}} = \pm \frac{8}{3}$ Differentiate equation with respect to x: $32x + 18y \left(\frac{dy}{dx} \right) = 0 \Rightarrow \frac{dy}{dx} = \frac{-32x}{18y} = -\frac{16x}{9y}$ $\frac{dx}{dt} = \frac{dx}{dy} \times \frac{dy}{dt} = -\frac{9y}{16x} \times 2 = -\frac{9y}{8x}$ For $\frac{dx}{dt} > 0$, x and y have different signs, and so the particle increases with respect to x at $(\sqrt{5}, -\frac{8}{3})$. [Alternative 1 for position of particle : Since $\frac{dy}{dt}, x > 0$, particle moves in anti-clockwise direction. Hence for $\frac{dx}{dt} > 0$, y should be negative.] [Alternative 2 for position of particle : differentiate w.r.t t and get $32x \frac{dx}{dt} + 18y \frac{dy}{dt} = 0$. Since $\frac{dx}{dt}, \frac{dy}{dt}, x > 0$, y should be negative.] At $(\sqrt{5}, -\frac{8}{3})$, $\frac{dx}{dt} = -\frac{9}{8} \times \frac{-8}{3\sqrt{5}} = \frac{3}{\sqrt{5}} \text{ cms}^{-1}$ At $(\sqrt{5}, -\frac{8}{3})$, its rate of increase is $\frac{3}{\sqrt{5}} \text{ cms}^{-1}$ [Alternative for $\frac{dx}{dt}$, $32x \frac{dx}{dt} + 18y \frac{dy}{dt} = 0 \Rightarrow 32\sqrt{5} \frac{dx}{dt} + 18\left(\frac{-8}{3}\right)(2) = 0 \Rightarrow \frac{dx}{dt} = \frac{3}{\sqrt{5}}]$

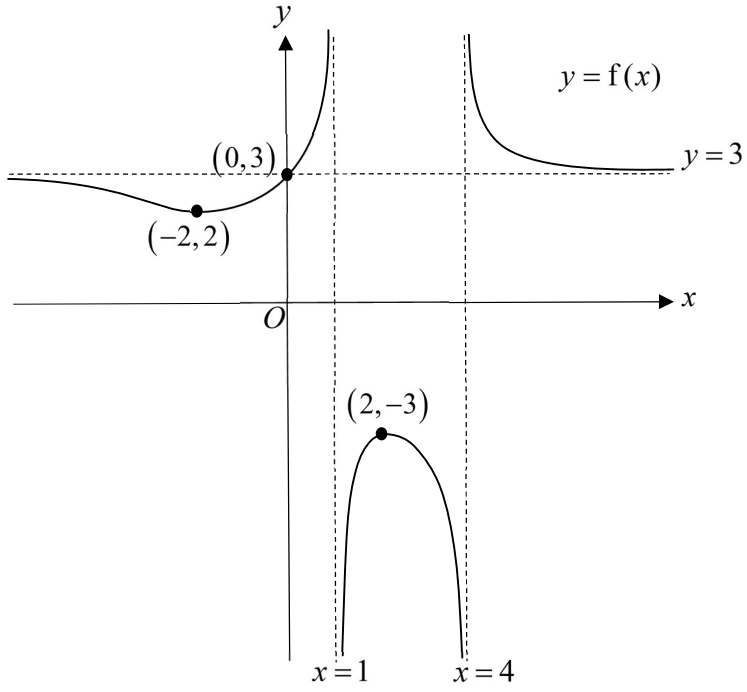
As the value of y is **negative** in this case, rejection of any answer would **need to be justified** using the conditions given in the question.

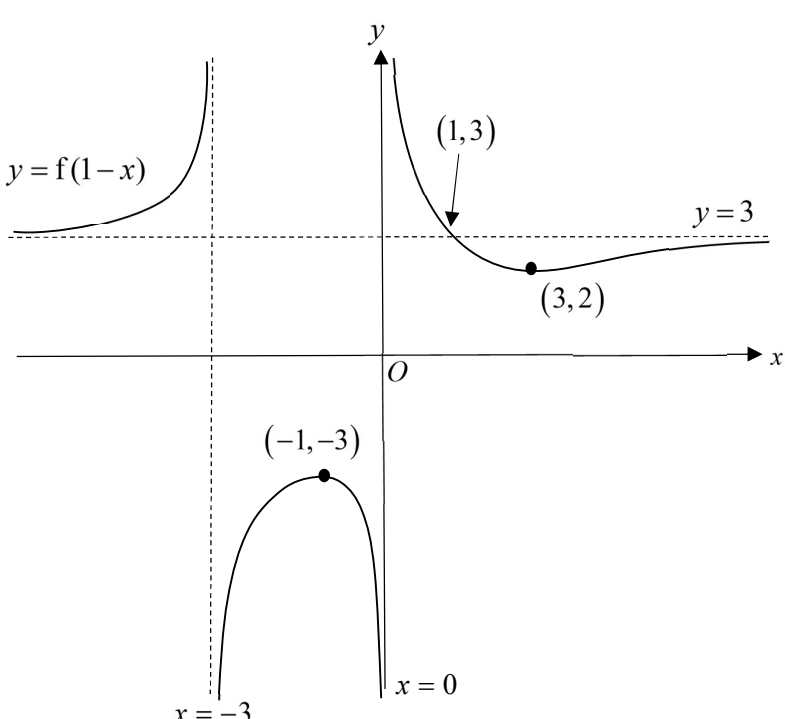
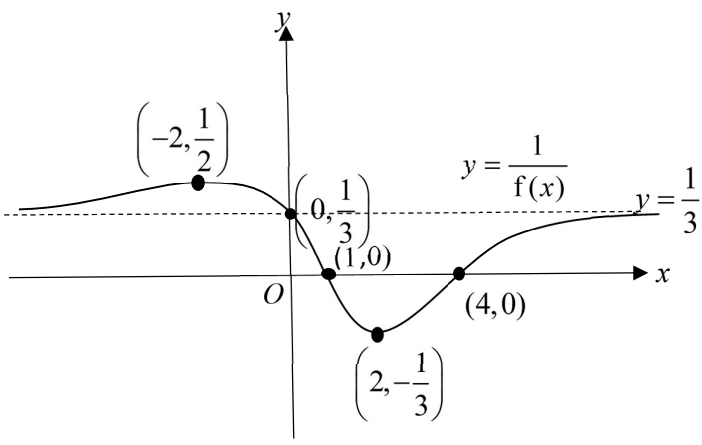
2	Mr. Li invested a total of \$30000 and divided this sum into three accounts, which paid 2%, 3% and 5% annual interest respectively. At the end of the first year, Mr. Li withdrew all the money out from the 2% and 5% accounts and gave the interest earned to his son. The amount in the 3% account, including interest, was re-invested in the same account for another year. At the end of the second year, Mr. Li withdrew all the money out of the 3% account and gave the interest earned to his son. The total interest received by the son from the three accounts was \$1423.50. Given that the amount invested in the 2% account was \$1000 more than the amount invested in the 5% account, find the amounts invested in each of the three accounts. [4]		
[4]	<table border="1"> <tr> <td data-bbox="332 630 1198 1008"> Let x, y and z be the amounts he invested into the 2%, 3% and 5% accounts respectively. $x + y + z = 30000 \text{ ----- (1)}$ $0.02x + (1.03^2 - 1)y + 0.05z = 1423.50 \text{ ----- (2)}$ $x - z = 1000 \text{ ----- (3)}$ From GC, solving the 3 equations, $x = 8000$, $y = 15000$, $z = 7000$ </td><td data-bbox="1198 630 1427 1008"> Student should try to identify this type of question as one that is easily solved with the GC using a system of three equations. </td></tr> </table>	Let x, y and z be the amounts he invested into the 2%, 3% and 5% accounts respectively. $x + y + z = 30000 \text{ ----- (1)}$ $0.02x + (1.03^2 - 1)y + 0.05z = 1423.50 \text{ ----- (2)}$ $x - z = 1000 \text{ ----- (3)}$ From GC, solving the 3 equations, $x = 8000$, $y = 15000$, $z = 7000$	Student should try to identify this type of question as one that is easily solved with the GC using a system of three equations.
Let x, y and z be the amounts he invested into the 2%, 3% and 5% accounts respectively. $x + y + z = 30000 \text{ ----- (1)}$ $0.02x + (1.03^2 - 1)y + 0.05z = 1423.50 \text{ ----- (2)}$ $x - z = 1000 \text{ ----- (3)}$ From GC, solving the 3 equations, $x = 8000$, $y = 15000$, $z = 7000$	Student should try to identify this type of question as one that is easily solved with the GC using a system of three equations.		
	A common mistake is that students computed the interest for the 3% account only for the second year ($0.03 \times 1.03y = 0.0309y$) and leaving out the interest for the first year ($0.03y$). Computing the interest earned over two years as $(1.03^2 - 1)y = 0.0609y$ would be a safer way.		

3	(i)	Prove that for $x > 0$, the substitution $y = ux$ reduces the differential equation $(y-x)\left(\frac{dy}{dx} - \frac{y}{x}\right) = y^2 + 2x^2$ to $\left(\frac{u}{u^2+2} - \frac{1}{u^2+2}\right)\left(\frac{du}{dx}\right) = 1. \quad [2]$
	(ii)	Hence find the general solution of the differential equation $(y-x)\left(\frac{dy}{dx} - \frac{y}{x}\right) = y^2 + 2x^2$ for $x > 0$. $[3]$
(i) [2]		$y = ux \text{ --- (1)}$ Differentiate w.r.t x, $\frac{dy}{dx} = x \frac{du}{dx} + u \text{ --- (2)}$ Substitute (1) and (2) into $(y-x)\left(\frac{dy}{dx} - \frac{y}{x}\right) = y^2 + 2x^2$ $\Rightarrow (ux-x)\left(x \frac{du}{dx} + u - u\right) = u^2x^2 + 2x^2$ $\Rightarrow (ux-x)\left(x \frac{du}{dx}\right) = u^2x^2 + 2x^2$ $\Rightarrow (u-1)\left(\frac{du}{dx}\right) = u^2 + 2 \quad \because x > 0$ $\Rightarrow \left(\frac{u-1}{u^2+2}\right)\left(\frac{du}{dx}\right) = 1$ $\Rightarrow \left(\frac{u}{u^2+2} - \frac{1}{u^2+2}\right)\left(\frac{du}{dx}\right) = 1 \text{ (shown)}$ Asked to reduce the DE to one with $\frac{du}{dx}$ so clearly u is a function of x, and cannot be a constant in $y = ux$
(ii) [3]		$\left(\frac{u}{u^2+2} - \frac{1}{u^2+2}\right)\left(\frac{du}{dx}\right) = 1$ Integrate w.r.t. x, $\int \frac{u}{u^2+2} - \frac{1}{u^2+2} du = \int 1 dx$ $x = \frac{1}{2} \ln(u^2+2) - \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{u}{\sqrt{2}}\right) + C$ $x = \frac{1}{2} \ln\left(\left(\frac{y}{x}\right)^2 + 2\right) - \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{y}{\sqrt{2}x}\right) + C$ Remember to replace u by $u = \frac{y}{x}$ for the final expression.

4	(i)	Sketch the curve with equation $x^2 + y^2 - 6x = 7$. [2]
	(ii)	The region R is bounded by the curve $x^2 + y^2 - 6x = 7$, for $x \geq 3$, and the line $x = 3$. Given that $\int_0^a \sqrt{a^2 - x^2} dx = \frac{\pi a^2}{4}$, find the exact volume of the solid of revolution formed when R is rotated completely about the y-axis. [4]
(i) [2]	$x^2 + y^2 - 6x = 7$ $(x-3)^2 + y^2 - 9 = 7$ $(x-3)^2 + y^2 = 16$ $(x-3)^2 + y^2 = 4^2$  Circle with centre (3, 0) and radius 4.	Use compass set to draw a circle. (use equal scale) Indicate clearly “critical features” like centre and radius.
(ii) [4]	$x = 3 \Rightarrow 9 + y^2 - 6(3) = 7 \Rightarrow y^2 = 16 \Rightarrow y = \pm 4$ $x^2 + y^2 - 6x = 7 \Rightarrow (x-3)^2 + y^2 = 16 \Rightarrow x = 3 \pm \sqrt{16 - y^2}$ Since $x \geq 3$, $x = 3 + \sqrt{16 - y^2}$. Volume of solid generated $= \pi \int_{-4}^4 x^2 dy - \pi (3)^2 (2(4))$ $= \pi \int_{-4}^4 \left(3 + \sqrt{16 - y^2}\right)^2 dy - 72\pi$ $= \pi \int_{-4}^4 \left(9 + 6\sqrt{16 - y^2} + 16 - y^2\right) dy - 72\pi$ $= \pi \left[25y - \frac{y^3}{3} \right]_{-4}^4 + (6\pi)(2) \left[\frac{\pi}{4} (4^2) \right] - 72\pi$ $= \pi \left(200 - \frac{128}{3} \right) + 48\pi - 72\pi$ $= \frac{256}{3} \pi + 48\pi$	R is rotated completely about the y-axis so the integral involved should be “...dy”. Note that volume of required region is to take “outer volume” subtract “inner volume” which is a cylinder.

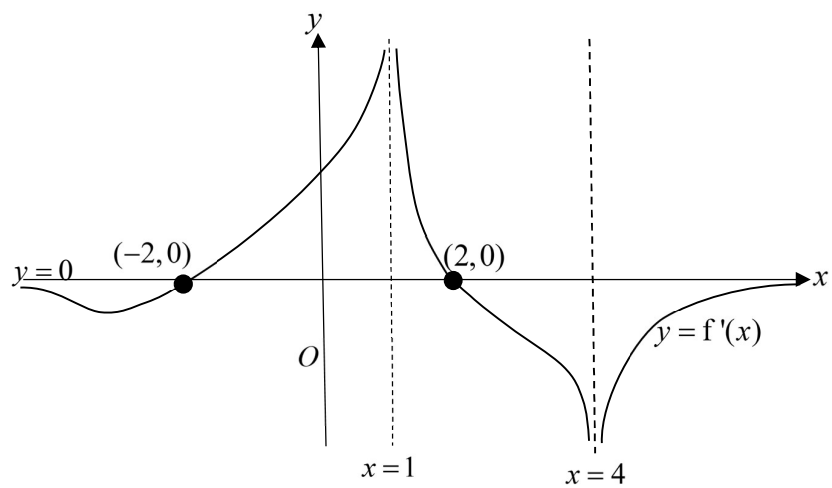
5	(a)	Find $\int x \cos x^2 dx$.	[1]
	(b)	Use integration by parts to find $\int x \cos 2x dx$.	[3]
		Hence or otherwise find $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} x \cos^2 x dx$.	[3]
(i) [1]		$\int x \cos x^2 dx = \frac{1}{2} \sin x^2 + c$	More generally, $\int f'(x) \cos(f(x)) dx$ $= \sin(f(x)) + c$ Don't forget "+ c"
(ii) [3]		$\int x \cos 2x dx = \frac{x}{2} \sin 2x - \frac{1}{2} \int \sin 2x$ $u = x \quad \frac{dv}{dx} = \cos 2x$ $\frac{du}{dx} = 1 \quad v = \frac{1}{2} \sin 2x$ $\int x \cos 2x dx = \frac{x}{2} \sin 2x + \frac{\cos 2x}{4} + c$	
[3]		$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} x \cos^2 x dx = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} x \left(\frac{\cos 2x + 1}{2} \right) dx$ $= \frac{1}{2} \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} x \cos 2x dx + \frac{1}{2} \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} x dx$ $= \frac{1}{2} \left[\frac{x}{2} \sin 2x + \frac{\cos 2x}{4} \right]_{\frac{\pi}{4}}^{\frac{\pi}{2}} + \frac{1}{2} \left[\frac{x^2}{2} \right]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$ $= \frac{1}{2} \left[\left(0 - \frac{1}{4} \right) - \left(\frac{\pi}{8} + 0 \right) \right] + \frac{1}{2} \left[\frac{1}{2} \left(\frac{\pi^2}{4} - \frac{\pi^2}{16} \right) \right]$ $= \frac{3\pi^2}{64} - \frac{1}{8} - \frac{\pi}{16}$	Double angle formula for cosine is given in MF27

6	The diagram below shows the curve of $y = f(x)$. The curve has a minimum point at $(-2, 2)$, a maximum point at $(2, -3)$ and cuts the y-axis at $(0, 3)$. The lines $x = 1$, $x = 4$ and $y = 3$ are the asymptotes to the curve.  On separate diagrams, draw sketches of the following graphs, stating the exact coordinates of any turning points and/or points of intersection with the axes, and the equations of any asymptotes, where possible.
(a)	$y = f(1 - x)$. [3]
(b)	$y = \frac{1}{f(x)}$. [3]
(c)	$y = f'(x)$. [3]
	Label axes and origin. Use a ruler to draw any lines. Write neatly and use a “dark pencil”. Please remove unwanted pencil markings cleanly with an eraser.

(a) [2]		
(b) [3]		The only asymptote is the line $y = \frac{1}{3}$. As such, there should not be other lines (such as $x = 1$ or $x = 4$) seen in the sketch.

(c)

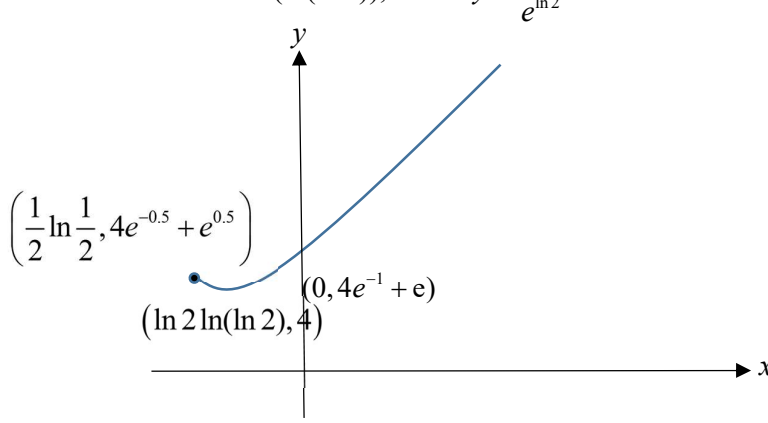
[3]



7	It is given that $\sum_{r=3}^n \frac{(3r-8)2^r}{r(r-2)} = \frac{(3n-2)2^{n+1}}{n(n-1)} - 16$.	
	Find $\sum_{r=1}^n \frac{(3r-2)2^r}{r(r+2)}$ in the form $\frac{(3n+A)2^{n+1}}{(n+B)(n+1)} + C$ where A, B and C are integers to be determined. [4]	
[4]	$\begin{aligned} \sum_{r=1}^n \frac{(3r-2)2^r}{r(r+2)} &= \sum_{r=3}^{n+2} \frac{(3(r-2)-2)2^{r-2}}{(r-2)r} \\ &= \sum_{r=3}^{n+2} \frac{(3r-8)2^{r-2}}{r(r-2)} \\ &= \frac{1}{4} \sum_{r=3}^{n+2} \frac{(3r-8)2^r}{r(r-2)} \\ &= \frac{1}{4} \left[\frac{(3(n+2)-2)2^{(n+2)+1}}{(n+2)(n+1)} - 16 \right] \\ &= \frac{(3n+4)2^{n+1}}{(n+2)(n+1)} - 4 \\ \therefore A &= 4, B = 2, C = -4 \end{aligned}$	Use the denominator ** $r(r+2)$ to suggest a replacement of r with “$r-2$” which appears in result in (i) Adjust lower and upper limit accordingly.

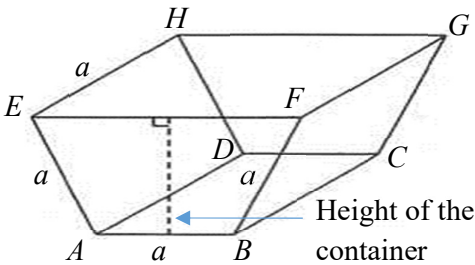
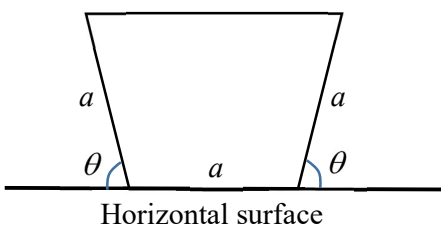
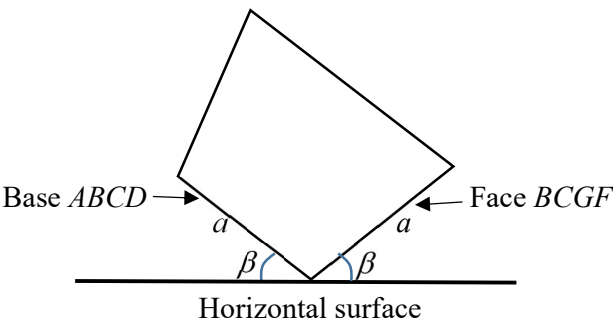
8	Do not use a calculator in answering this question.		
(a)	(i)	Solve the equation $z^2 = 4i - 3$.	[3]
	(ii)	Solve the equation $z^4 + 6z^2 + 25 = 0$.	[3]
(b)	Find the modulus and argument of the complex number $w = \frac{8-2i}{5+3i}$. Hence find the possible values of the positive integer n for which w^n is real.		
(a)	Let $z = x + iy$, $x, y \in \mathbb{R}$. Then		
(i)	$z^2 = 4i - 3 \Rightarrow (x + iy)^2 = (x^2 - y^2) + 2ixy = 4i - 3$		
[3]	$\Rightarrow \begin{cases} x^2 - y^2 = -3 \\ 2xy = 4 \end{cases}$		
	$\Rightarrow x^2 - \frac{4}{x^2} = -3$		
	$\Rightarrow x^4 + 3x^2 - 4 = (x^2 + 4)(x^2 - 1) = 0$		
	$\Rightarrow x = \pm 1$		
	When $x = 1, y = 2$. When $x = -1, y = -2$		
	Thus the roots are $1 + 2i$ and $-1 - 2i$.		
(a)	$z^4 + 6z^2 + 25 = 0 \quad \text{--- (1)} \quad z^4 + 6z^2 + 25 = 0 \quad \text{--- (1)}$		
(ii)	$(z^2 + 3)^2 + 16 = 0 \quad \text{or} \quad z^2 = \frac{-6 \pm \sqrt{36 - 4(25)}}{2}$		
[3]	$z^2 + 3 = \pm 4i$		
	$z^2 = 4i - 3 \quad \text{or} \quad -4i - 3 \quad = -3 \pm \frac{8i}{2} = -3 \pm 4i$		
	For $z^2 = 4i - 3$, $z = 1 + 2i, -1 - 2i$		
	Since (1) is an equation with real coefficients, the roots occur in conjugate pairs. Thus the roots of the equation $z^4 + 6z^2 + 25 = 0$ are $z = 1 \pm 2i, -1 \pm 2i$.		
(b)	$w = \frac{8-2i}{5+3i} = \frac{8-2i}{5+3i} \times \frac{5-3i}{5-3i}$		
[4]	$= \frac{40 - 10i - 24i + 6i^2}{5^2 + 3^2}$		
	$= \frac{34 - 34i}{34} = 1 - i$		
	Thus $ w = \sqrt{1^2 + 1^2} = \sqrt{2}$ and $\arg w = -\frac{\pi}{4}$		
	For w^n to be real, $w^n = (\sqrt{2})^n \left[\cos\left(-\frac{n\pi}{4}\right) + i \sin\left(-\frac{n\pi}{4}\right) \right]$ is real.		
	Hence $\sin\left(-\frac{n\pi}{4}\right) = 0 \Rightarrow \frac{n\pi}{4} = k\pi, k \in \mathbb{Z}$, and so		
	$n = 4k, k \in \mathbb{Z}^+$ (since $n > 0$)		
A calculator is “not allowed” for this question so show detailed working (you can check answers using calculator though).			
The set of “positive integer” does not include 0.			
Also, a general statement that n is a multiple of 4 is incorrect as this includes negative numbers.			

9	A curve C has parametric equations $x = t \ln t, \quad y = \frac{4}{e^t} + e^t,$ for $t \geq \frac{1}{2}$.	
	(i) C meets the y-axis at point P and line L is the normal to C at P. Show that the equation of L is $y = \frac{e}{4 - e^2}x + \frac{4 + e^2}{e}. \quad [4]$	
	(ii) Sketch the curve C, stating the coordinates of any turning points and points intersection with the axes. [3]	
	(iii) The finite region bounded by C, L and the line $x = \frac{1}{2} \ln \left(\frac{1}{2} \right)$ is denoted by R. Find the area of R. [3]	
(i) [4]	$x = t \ln t \Rightarrow \frac{dx}{dt} = t \left(\frac{1}{t} \right) + \ln t = 1 + \ln t,$ $y = \frac{4}{e^t} + e^t \Rightarrow \frac{dy}{dt} = -4e^{-t} + e^t = \frac{e^{2t} - 4}{e^t},$ $\therefore \frac{dy}{dx} = \frac{e^{2t} - 4}{e^t(1 + \ln t)}$ Now, $x = 0 \Rightarrow t \ln t = 0 \Rightarrow t = 1$ ($\because t > 0$) $\Rightarrow y = \frac{4}{e} + e = \frac{4 + e^2}{e} \quad \text{and} \quad \frac{dy}{dx} = \frac{e^2 - 4}{e}$ Equation of normal at $P \left(0, \frac{4 + e^2}{e} \right)$: $y - \frac{4 + e^2}{e} = -\frac{e}{e^2 - 4}x \Rightarrow y = \frac{e}{4 - e^2}x + \frac{4 + e^2}{e}$	As this is a “show ...” question, detailed working should be made clear.
(ii) [3]	$\frac{dy}{dx} = \frac{e^{2t} - 4}{e^t(1 + \ln t)} = 0$ $\Rightarrow e^{2t} - 4 = 0$ $\Rightarrow t = \ln 2$	For parametric curves, it is important to note the range of t to sketch. Use the ZOOM function in the GC to check the exact shape/feature of the curve. Graphs should be drawn sufficiently large so that the key features (turning

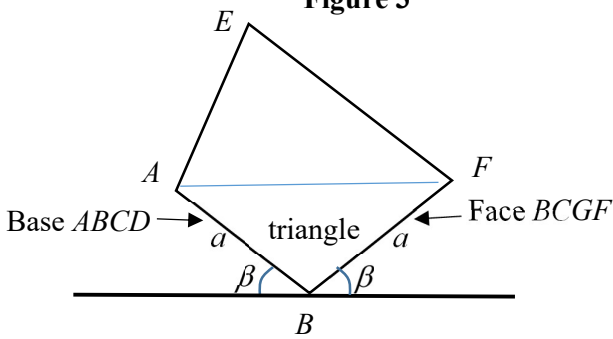
	Min occurs at $x = \ln 2(\ln(\ln 2))$, $y = \frac{4}{e^{\ln 2}} + e^{\ln 2} = 4$  $\left(\frac{1}{2} \ln \frac{1}{2}, 4e^{-0.5} + e^{0.5}\right)$ $(\ln 2 \ln(\ln 2), 4)$ $(0, 4e^{-1} + e)$	points and relative position) can be seen clearly. Where exact values is not stated, students may indicate the coordinates in 3 s.f. . the y values of the turning point, end-point and y-intercept are 4, 4.07 and 4.19, respectively, so the relative position should be indicated clearly in the diagram.
(iii) [4]	Area $= \int_{0.5 \ln 0.5}^0 \left(\frac{e}{4 - e^2} x + \frac{4 + e^2}{e} \right) - y \, dx$ $= \int_{0.5 \ln 0.5}^0 \left(\frac{e}{4 - e^2} x + \frac{4 + e^2}{e} \right) dx - \int_{\frac{1}{2}}^1 \left(\frac{4}{e^t} + e^t \right) (1 + \ln t) \, dt$ = 0.0943 (3s.f.)	For questions involving Area, draw the respective curves and identify the required region, and use the idea of ‘Area under Upper curve – Lower Curve’ ($\int_a^b y_1 - y_2 \, dx$ in this case) Be familiar with how to find area of parametric curves, by using “substitution”.

10	The plane π_1 contains the point $A(1, 2, -1)$ and the line l with equation $\frac{x-1}{2} = \frac{2-z}{3}, y = -1$. The plane π_2 contains the point $B(-5.5, 3, 2)$ and meets π_1 in the line l .		
	(i)	Find the equation of π_1 in scalar product form. [3]	
	(ii)	Show that the vector $\overrightarrow{BF}$ is $\begin{pmatrix} 4.5 \\ -4 \\ 3 \end{pmatrix}$, where F is the foot of perpendicular from B to l . [3]	
	(iii)	Find the exact value of the shortest distance from B to π_1 . [2]	
	(iv)	Hence, or otherwise, find the acute angle between π_1 and π_2 , giving your answer to the nearest 0.1° . [3]	
(i) [3]	$l: \mathbf{r} = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 0 \\ -3 \end{pmatrix}, \lambda \in \mathbb{R}$ Let C be a point on l such that $\overrightarrow{OC} = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$. $\overrightarrow{AC} = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ -3 \\ 3 \end{pmatrix}$ $\mathbf{n}_1 = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \times \begin{pmatrix} 2 \\ 0 \\ -3 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix}$ $\pi_1: \mathbf{r} \cdot \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix} = 3 - 2 + 4 = 5 \Rightarrow \mathbf{r} \cdot \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix} = 5$		Use proper vector notation: Put the tilde beneath the $\mathbf{r}$. Write "... $\lambda \in \mathbb{R}$ " in the equation of a line.
(ii) [3]	$\overrightarrow{OF} = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 0 \\ -3 \end{pmatrix}, \text{ for some } \lambda \in \mathbb{R}$ $\overrightarrow{BF} = \overrightarrow{OF} - \overrightarrow{OB} = \begin{pmatrix} 6.5 + 2\lambda \\ -4 \\ -3\lambda \end{pmatrix}$		

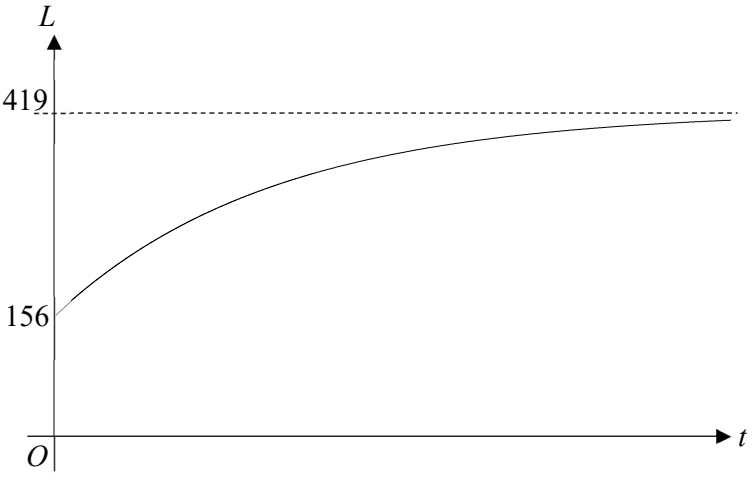
	$\overrightarrow{BF} \cdot \begin{pmatrix} 2 \\ 0 \\ -3 \end{pmatrix} = 0 \Rightarrow 13 + 4\lambda + 9\lambda = 0$ $\lambda = -1$ $\therefore \overrightarrow{BF} = \begin{pmatrix} 6.5 + 2(-1) \\ -4 \\ -3(-1) \end{pmatrix} = \begin{pmatrix} 4.5 \\ -4 \\ 3 \end{pmatrix} \text{ (shown)}$	As this is a “show ...” question, show the substitution of $\lambda = -1$
(iii) [2]	Shortest distance from B to π_1 = length of projection of $\overrightarrow{BF}$ onto $\mathbf{n}_1$ $= \overrightarrow{BF} \cdot \hat{\mathbf{n}}_1 = \frac{\begin{pmatrix} 4.5 \\ -4 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix}}{\sqrt{17}} = \frac{23}{2\sqrt{17}} \text{ units}$	
(iv) [3]	Let θ be the acute angle between π_1 and π_2 $\sin \theta = \frac{23}{BF}$ $= \frac{23}{2\sqrt{17}} \div \overrightarrow{BF} $ $\theta = \sin^{-1} \frac{23}{2\sqrt{17}\sqrt{4.5^2 + 4^2 + 3^2}} = 24.5^\circ \text{ (1d.p)}$ [Alternatively, $\mathbf{n}_2 = \overrightarrow{BF} \times \begin{pmatrix} 2 \\ 0 \\ -3 \end{pmatrix} = \begin{pmatrix} 4.5 \\ -4 \\ 3 \end{pmatrix} \times \begin{pmatrix} 2 \\ 0 \\ -3 \end{pmatrix} = \begin{pmatrix} 12 \\ 19.5 \\ 8 \end{pmatrix}$ Let θ be the acute angle between π_1 and $\frac{23}{2\sqrt{17}}$ $\theta = \cos^{-1} \frac{\begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 12 \\ 19.5 \\ 8 \end{pmatrix}}{\sqrt{17}\sqrt{588.25}} = \cos^{-1} \frac{91}{\sqrt{17}\sqrt{588.25}} = 24.5^\circ \text{ (1d.p)]}$ [Another alternative method was to find the acute angle between $\overrightarrow{BF}$ and $\mathbf{n}_1$ using the dot product formula, then subtract it from 90°.]	

11	Figure 1 shows an open container in the form of a trapezoidal prism $ABCDEFGH$ with square base $ABCD$ and $AB = AE = BF = EH = a$ cm, where a is a constant. The container is made of plastic of negligible thickness and is placed on a horizontal surface. The faces $BCGF$ and $ADHE$ are inclined at an angle θ radians, $0 < \theta < \frac{\pi}{2}$, to the horizontal, and faces $ABFE$ and $DCGH$ are perpendicular to base $ABCD$. Figure 2 shows its cross-sectional view. Figure 1  Trapezoidal Prism Figure 2  Cross-Sectional View
(i)	Show that the volume V cm³ of the container is given by $V = a^3 \sin \theta (1 + \cos \theta)$. [2]
(ii)	Use differentiation to find, in terms of a, the maximum value of V in exact form, proving that it is a maximum. [5]
(iii)	A container is constructed with $\theta = \frac{\pi}{3}$ and it is filled with water to half its height. Find, in terms of a, the exact volume of water in the container. [3] The container is then tilted in the direction of the face $BCGF$ until face $BCGF$ and base $ABCD$ makes the same angle β radians, where β is a constant, with the horizontal. Figure 3 shows its cross-sectional view. Figure 3  Cross-Sectional View Explain if it is possible to tilt the container to this position without any water flowing out from the container. [2]

(i) [2]	By trigo ratio, $EX = a \sin \theta$, $AX = a \cos \theta$ $V = a \left\{ 2 \left[\frac{1}{2} (a \cos \theta)(a \sin \theta) \right] + a(a \sin \theta) \right\}$ $= a^3 \sin \theta (1 + \cos \theta)$ [Alternatively, by area of trapezium $V = a \left\{ \frac{1}{2} (a \sin \theta) [a + (a + 2a \cos \theta)] \right\} = a^3 \sin \theta (1 + \cos \theta)]$	A “show” question so there should be clear explanation in working or illustration on the diagram indicating the method used.
(ii) [5]	$\frac{dV}{d\theta} = a^3 [\cos \theta (1 + \cos \theta) + \sin \theta (-\sin \theta)]$ $= a^3 [\cos \theta + \cos 2\theta] \quad (\text{Double angle formulae})$ $= \begin{cases} 2a^3 \cos \frac{3\theta}{2} \cos \frac{\theta}{2} & (\text{Factor Formulae}) \\ a^3 [2 \cos^2 \theta + \cos \theta - 1] & (\text{Double angle formulae}) \\ a^3 (2 \cos \theta - 1)(\cos \theta + 1) \end{cases}$ $\frac{dV}{d\theta} = 0 \Rightarrow \cos \frac{3\theta}{2} = 0 \text{ or } \cos \frac{\theta}{2} = 0$ $\Rightarrow \frac{3\theta}{2} \text{ or } \frac{\theta}{2} = \frac{\pi}{2}$ $\Rightarrow \theta = \frac{\pi}{3} \text{ or } \pi \text{ (NA)}$ $\left[\begin{array}{l} \frac{dV}{d\theta} = 0 \Rightarrow \cos \theta = \frac{1}{2} \text{ or } \cos \theta = -1 \\ \Rightarrow \theta = \frac{\pi}{3} \text{ or } \pi \text{ (NA)} \end{array} \right]$ $\frac{dV}{d\theta} = a^3 [\cos \theta + \cos 2\theta]$ At $\theta = \frac{\pi}{3}$, $\frac{d^2V}{d\theta^2} = a^3 [-\sin \theta - 2 \sin 2\theta]$ $= a^3 \left[-\frac{\sqrt{3}}{2} - \sqrt{3} \right] = -\frac{3\sqrt{3}}{2} a^3 < 0$ Hence $\theta = \frac{\pi}{3}$ gives maximum value of V and $\max V = a^3 \left(\frac{\sqrt{3}}{2} \right) \left(1 + \frac{1}{2} \right) = \frac{3\sqrt{3}}{4} a^3 \text{ cm}^3$	For students who use first derivative test, clearly state what are the signs of the 2 factors of $\frac{dV}{d\theta}$ when $\theta = \frac{\pi^-}{3}, \frac{\pi^+}{3}$ and thus the final sign, to conclude if the gradient is positive or negative. Note that the conclusion is for the maximum value of V, and not of θ

(iii) [3+2]	Half its height = $\frac{1}{2} \left(a \sin \frac{\pi}{3} \right) = \frac{\sqrt{3}a}{4}$. $\therefore \tan \frac{\pi}{3} = \frac{\frac{\sqrt{3}a}{4}}{x} \Rightarrow \sqrt{3} = \frac{\sqrt{3}a}{4x} \Rightarrow x = \frac{a}{4}$ $V(\text{half its height})$ $= a \left[\left(\frac{\sqrt{3}a}{4} \right) a + \left(\frac{a}{4} \right) \left(\frac{\sqrt{3}a}{4} \right) \right] = \left(\frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{16} \right) a^3 = \frac{5\sqrt{3}}{16} a^3 \text{ cm}^3$ Figure 3  Cross-Sectional View Note that we can consider face $ABFE$ as a possible cross-sectional view in Figure 3. Then $\angle ABF = \frac{2\pi}{3}$ and $AB = BF = a$, and so Area of triangle $ABF = \frac{1}{2} a^2 \sin \frac{2\pi}{3} = \frac{\sqrt{3}}{4} a^2$. Hence the volume of water that the container can hold at this position is at most $\frac{\sqrt{3}}{4} a^3 < V(\text{half the height})$, and so water will definitely flow out of the container before it reaches this position. So no, it is not possible. [Alternative explanation (for the case where θ may not be fixed): Note that area of triangle increases with θ and θ is acute. Hence max volume $< \frac{a^3}{2} \sin \frac{\pi}{2} = \frac{a^3}{2} < \frac{5\sqrt{3}}{16} a^3$.]	Students should explain their working clearly, such as using similar triangles etc. Please remove all unwanted or replaced working and not leave multiple methods.
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12	The von Bertalanffy growth model, introduced by von Bertalanffy in 1938, is a widely used growth curve and is especially important in fisheries studies. It is used to predict the length, L mm of a fish over a period of time, t years. If L_{∞} is the maximum length for a species, then the model assumes that the rate of growth in length of a fish is proportional to $L_{\infty} - L$.		
	(i)	By setting up and solving a differential equation, show that the general solution of this differential equation is given by $L = L_{\infty} - Ae^{-kt}$, where k is the constant of proportionality and A is a positive constant. [5]	
	For the species of fish known as the Atlantic croaker, it has been determined that $L_{\infty} = 419$ mm and at one year of age, its length is 219 mm and the rate of growth in length is 55 mm per year. Using the above model, obtain an expression for L in terms of t . [3]		
	(ii)	Find its age when the Atlantic croaker grows to a length of 300 mm. [2]	
	(iii)	Sketch a graph of L against t . [2]	
(i) [5]	$\frac{dL}{dt} = k(L_{\infty} - L)$, where k is the constant of proportionality. $\frac{dL}{dt} = k(L_{\infty} - L)$ $\frac{1}{L_{\infty} - L} \frac{dL}{dt} = k$ $\int \frac{1}{L_{\infty} - L} dL = \int k dt$ $-\ln(L_{\infty} - L) = kt + c \qquad \because L_{\infty} - L > 0$ $L_{\infty} - L = e^{-(kt+c)}$ $L_{\infty} - L = Ae^{-kt}, \quad A \text{ is a positive constant}$ $L = L_{\infty} - Ae^{-kt}, \quad A \text{ is a positive constant}$ Note that $L \neq L_{\infty}$ in this context. Also, $A > 0$.		A “show ...” question so all steps must be clearly explained including why “ A ” must be positive
[3]	Since $L_{\infty} = 419$ mm, $L = 419 - Ae^{-kt}$ When $t = 1$, $L = 219$ and thus $219 = 419 - Ae^{-k} \Rightarrow Ae^{-k} = 200 \text{ --- (1)}$ Also, $t = 1$, $\frac{dL}{dt} = 55 \Rightarrow Ake^{-k} = 55 \text{ --- (2)}$		

	Sub (1) into (2), $k = \frac{55}{200} = \frac{11}{40}$ (or 0.275) (alternatively, using $\frac{dL}{dt} = k(L_{\infty} - L)$, $55 = k(419 - 219)$) Thus $A = 200(e^{\frac{11}{40}}) = 263.31 = 263$ $L = 419 - 263e^{-\frac{11}{40}t}$	
(ii) [2]	When $L = 300$, $300 = 419 - 263.31e^{-\frac{11}{40}t} \Rightarrow t = 2.89$ years	
(iii) [2]		Draw horizontal asymptote.